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Combustion and Flame 151 (2007) 412–425
Combustion and Flame
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Combustion dynamics of a low-swirl combustor

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Received 28 September 2006; received in revised form 20 April 2007; accepted 13 July 2007

Available online 24 September 2007

Abstract

The methodology for the measurement of dynamic combustor behavior has never been clearly established, due to the complexities associated with unsteady premixed flames and the difficulties in their measurement. The global and local distribution of Rayleigh index and the flame response functions are the main parameters normally employed to quantify and describe combustion dynamics. The Rayleigh index quantifies the thermoacoustic coupling, while the flame response function is a measure of the response of the system to outside disturbances. The primary objective of this work is to investigate the combustion dynamics of a commonly used low-swirl burner and to develop tools and methods for examining the dynamics of a combustion system. To this end, the effect of acoustic forcing at various frequencies on flame heat release behavior has been investigated. The current work uses OH–PLIF imaging of the flame region to produce phase-resolved measurements of flame behavior at each frequency. The response of the flame to the imposed acoustic field over the range of 22–400 Hz is then calculated from the processed images. This provides a starting point for an extension/extrapolation to practical acoustic ranges (∼5000 Hz). It was found that the thermoacoustic coupling was mainly evident in the shear mixing zone, producing a toroidal Rayleigh index distribution pattern. The phase shift of the flame fluctuation from the imposed acoustic wave seems to be very closely coupled to the vortices generated at the flame boundary due to shear mixing (Kelvin–Helmholtz instability), thus inducing the alternating toroidal structures. The peak value of the flame response function coincides with the peak absolute value of the Rayleigh index.
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Keywords: Combustion instability; OH; PLIF; Swirl; Combustion dynamics; Acoustically excited

1. Introduction

Combustion instabilities are pressure oscillations that are caused by coupling between the thermoacoustics and the fluid dynamics during a combustion processes. The interaction between vortices (which control mixing), sound waves, and combustion heat release can lead to self-excited oscillations that can cause problems such as reduced combustion efficiency, increased noise, and possible structural damage. Recently, environmental concerns have led to the use of lean premixed (LP) combustion schemes that reduce the amount of NO_x produced. The LP scheme achieves lower NO_x formation by reducing the fuel flow rate and thereby reducing the peak flame temperature. Unavoidably, these lean-burn conditions result in the occurrence of instabilities, since combustion in

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doi:10.1016/j.combustflame.2007.07.017

these systems tends to occur near the lean blow-out (LBO) limit [1], which is more susceptible to unstable combustion dynamics. Nair and Lieuwen [2] developed a diagnostic technique to monitor the combustor near LBO using the acoustic characteristics of the flame. This was an effort to utilize thermoacoustic measurements to determine how close the conditions were to LBO, in order to develop actively controlled systems that avoid LBO [3].

Most experimental work characterizing combustion instability has been performed on naturally unstable systems. A study of the acoustic coupling between fuel injectors and an applied acoustic field was carried out by Torger et al. [4], but it only considers cold-flow experiments. Work by Chen et al. [5] with premixed flames was specifically designed to simulate solid rocket propellants. Results were limited to one spatial dimension and only two forcing frequencies were used. The study by Cadou et al. [6] employed a two-dimensional dump combustor that showed little response to nonresonant forcing. Durox et al. [7] and Schuller et al. [8] measured the flame response to forced acoustic oscillations at frequencies up to 400 Hz, but the burner was placed in the open air, and the response function was based on the pressure and velocity measurements. They measured the CH chemiluminescence without attempting any phase resolution. Despite these efforts, work is still required to understand the physical underpinnings of this process and to provide industry with guidelines that will be useful in designing stable combustion systems.

In works by Dowling et al. [9], Bloxsidge et al. [10], and Langhorne [11], a general theoretical model was constructed based on experimental measurements of the "reheat buzz," which is a combustion instability observed in a combustor with confined geometry. These studies laid the foundation for the next step—to use the data as the fundamental reference for theoretical modeling. However, these experiments did not include a forced acoustic excitation, which is crucial in determining the dynamic response of the system. Also, the lack of spatial resolution in previous measurements means that the resulting data are primarily useful only for one-dimensional models, which are too simplistic for many of the issues of interest.

Previous measurements of flame intensity (chemiluminescence from all species) has been used for both control and modeling purposes. It is based on a line-of-sight measurement rather than a spatially resolved local measurement. Since radiation may be absorbed, the final intensity at the observation point does not in general represent the local reaction rate. Pun et al. [12], showed that misleading results can be obtained from chemiluminescence data.

Chemiluminescence specifically from the CH radical has been used to mark the reaction zone (although it suffers from line-of-sight limitations) and has been used by a number of researchers to study heat release in unsteady flames. These studies can be categorized into two groups: measurements using a photomultiplier tube with a slit that obscures a portion of the flame to obtain some spatial resolution, typically in the axial direction [5,13,14], and fully two-dimensional imaging using CCD-based cameras [1,15,16]. Of these works, only Chen et al. [5] acoustically forced a flame, but they used a photomultiplier tube with a slit configuration that obtained only one-dimensional information.

One of the first demonstrations of planar laser-induced fluorescence (PLIF) of the hydroxyl radical (OH) in a flame was performed by Dyer and Crosley [17]. PLIF has been used to identify a variety of species in unsteady reacting flows, including OH (to measure the heat release rate [18,19]) and NO that was seeded into the fuel in order to measure the temperature field [6]. Cadou et al. [6] and Pun et al. [12] applied OH–PLIF to achieve spatial resolution in their investigation of the dynamics of a forced combustion system.

Najm et al. [20] showed that the concentration of OH is not directly correlated with the heat release rate. However, for the flame considered in the present study (described below), and for the case of small perturbations to local flame intensity, there is reason to believe that use of the OH–LIF signal is an appropriate way to quantify heat release rates. This belief is due to (1) the assumption that small changes in the OH–LIF signal are proportional to the small changes in local heat release and (2) the assumption that the data processing involved in calculating the Rayleigh index uses only the changes in the OH–LIF signal that occur in the flame zone and not the steady post-flame signal. The logic of Part 1 is as follows: the intensity of the OH–LIF signal is proportional to the fluorescence quantum yield, which in turn is proportional to the number density of the OH radical. The local number density is proportional to the reciprocal of the local temperature. Therefore the spatial distribution of the OH–LIF signal is representative of the local heat release, since the OH–LIF signal is stronger where the local temperature is higher. The analysis method (Part 2) is based on calculating the spatially fixed temporal fluctuation in the OH signal. The post-flame regions are unchanged on the time scale of the pressure fluctuation used to identify changes in flame intensity/heat release. This means that the Rayleigh index will have a value of zero for all of the purely thermally produced OH regions.

Building on the efforts of Pun and co-workers with OH–PLIF measurements, the current series of experi-

ments considers the acoustic parameters to examine a wider range of conditions. Pun's experiments looked at system forcing at five frequencies (22, 27, 32, 37, 55 Hz). By forcing over a range of frequencies, data for transfer functions can be obtained and truly informative results for the combustion dynamics can be extracted. The current work expands to a wider range of forcing frequencies (up to 400 Hz) and employs the swirl burner geometry that is commonly used in industry.

The theoretical foundation for this work was developed by Culick [21,22], who defined a Rayleigh index that is based on Rayleigh's statements concerning acoustic waves and flame behavior. The first measurement of a spatially resolved Rayleigh index was reported by Pun et al. [12], who examined the coupled effects of acoustic forcing with combustion heat release using OH–PLIF and chemiluminescence. This revealed that in the flame region the response of the combustion process under acoustic excitations is strongly phase-dependent, and this dependence was quantified. The flame base behavior and NO production rates were also studied [23]. The present work uses phase-resolved OH–PLIF measurement of the flame region to measure flame behavior at each frequency in a spatially resolved manner. The response of the flame to the imposed acoustic field over the range of 22–400 Hz is then determined, to provide data that are expected to serve as a starting point for combustion system designs and for numerical estimations of combustor behavior up to the higher practical levels (∼5000 Hz). OH–PLIF is believed to give the best temporal and spatial information available at the present time, and it is superior to chemiluminescence techniques [12]. The current work is limited to 400 Hz due to experimental limitations, but it is expected that the flame response will decrease above 400 Hz and the trends observed at low frequencies are expected to yield insight about the trends expected at higher frequencies.

The objective of this work is to determine the global and local values of the fluctuations of local energy release rate (based on the OH–PLIF measurements) and to demonstrate the effectiveness of the experimental methodology. The measurements provide the basis for constructing a model of the source term $\dot{Q}'_{\mathrm{r}}$, which is a primary parameter in combustion dynamics models (Culick [21]). By "primary parameter," it is meant that $\dot{Q}'_{\mathrm{r}}$ appears as a source of unsteady motions in the equations for flow in a combustor and therefore can be given a direct physical interpretation. An accurate representation of $\dot{Q}'_{\mathrm{r}}$ is essential to understanding the presence of combustion instabilities and, accordingly, how they can be either avoided or eliminated in practical systems. In general, $\dot{Q}'_{\mathrm{r}}$ must be known to analyze any problem of combustor dynamics. To examine the impact of other key parameters on this problem, the behavior of flames at different equivalence ratios is also compared. Three different equivalence ratios (0.5, 0.6, and 0.75) were used, which correspond to leaner, desired, and richer operation conditions for the swirl burner, respectively. All three values of equivalence ratio are within the normal operating range of the burner.

2. Experimental configurations

2.1. Laser system

The experimental layout is shown in Fig. 1. The PLIF system employs an Nd:YAG pulsed (10-Hz) laser (Continuum Powerlite 9010) pumping a tunable dye laser (Continuum ND6000), which in turn drives a custom-made mixer/doubler system. The 532-nm beam out of the doubler pumps the dye laser. Use of Rhodamine 590 for the dye laser dissolved in methanol optimizes the conversion efficiency near 564 nm (∼200 mJ/pulse), which is then doubled to approximately 282 nm to excite the A–X (1, 0) band of OH. Slightly over 35 mJ/pulse of energy is provided by this system, resulting in saturated fluorescence.

2.2. Detection system

The detector for the fluorescence signal is an intensified CCD camera (Princeton Instruments), which uses a 512 × 512 Thomson CCD array, operated with a gate width of 100 ns. Attached to the camera is a catadioptric, all-reflective f/1.2 UV lens with a focal length of 105 mm, providing a spatial resolution of 174 μm. The lens provides exceptionally high light throughout, as well as minimizing spherical and chromatic aberrations. A lattice electro optics (LEO) bandpass filter (312.6F10-10) was installed in the ICCD camera to cut off all undesired light, such as that from the scattered pump laser beam, while transmitting the fluorescence that occurs at 300–320 nm. One thousand eight hundred images were taken for each set (single equivalence ratio) of data and these were phase-binned to give a phase-resolution of 10°.

2.3. Postprocessing

Since the images are taken at random values of the phase angle of the driving pressure wave, the camera gating signal is recorded by the data acquisition system along with the pressure signal to enable appropriate postprocessing. Postprocessing involves sorting of the images according to the phase of imposed

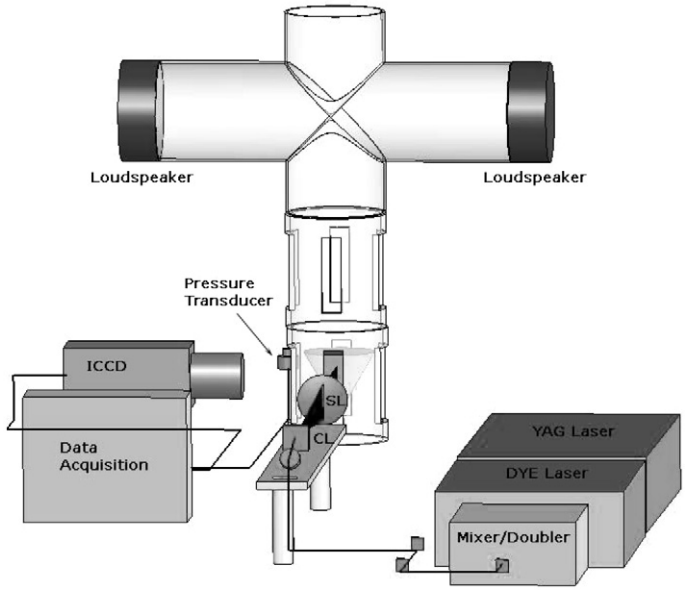

Fig. 1. Combustion chamber and layout of the laser and optics.

acoustic oscillations, creating phase-averaged images, and normalizing based on shot-to-shot laser intensity and other quantities.

Each image is tagged with its phase bin information, fluctuating pressure, and mean pressure values. After images are sorted and classified according to the corresponding phase information, quantities of interests such as the Rayleigh index, the heat release rate fluctuations, and the local/global flame responses are calculated.

2.4. Chamber and acoustics

A schematic view of the acoustic chamber is shown in Fig. 1. The acoustic chamber is made of an aluminum central section and a stainless steel top, with the top housing the loudspeakers (Cerwin-Vega, Vega 124) used for system excitation and enabling direct exhaust of the product gases. The exhaust port is open to the atmosphere which creates an acoustically open exit condition for acoustic wave resonance. The bottom of the chamber creates a closed-end acoustic condition. The speakers are 30 cm in diameter and powered by a 1000-W power amplifier (Mackie M1400i). A function generator (Wavetek 171) provides the input signal. The speakers create pressure and velocity oscillations in the combustion chamber (Fig. 1) with a peak-to-peak pressure inside the chamber of $0.1 \pm 0.01\%$ of the atmospheric pressure.

The amplitude of the fundamental driving mode (the acoustic waves) is actively controlled by custom-designed electronics, which measure the pressure in the acoustic chamber at the burner height with a pressure transducer (PCB 106B50) and then scale the power provided to the speakers. The signal from the transducer is notch-filtered to ensure that the intended driving mode is correctly amplified or attenuated.

2.5. Fuel/air feed system

A Linde air-flow meter/controller unit (18C-202-4125; ∼50 slpm) and a Hastings flow meter/controller unit (HFC-203A; ∼300 slpm) were used to provide a steady flow of air and methane to the premixer. The equivalence ratios, $\phi = 0.5$, 0.6, and 0.75, were selected based on the recommendation of the designer (Dr. Robert Cheng, Lawrence Berkeley Livermore Laboratory). The total volume flow rate is about 120 slpm, which corresponds to an exit velocity of 4 m/s out of the nozzle (see Table 1 for experimental details).

2.6. Swirl burner

The burner, shown in Figs. 2 and 3, is a swirl-stabilized burner with a swirl number of around 0.7. The inner diameter of the nozzle is 2.54 cm with the swirler located 3.81 cm below the burner exit. The swirler has a small grid at the center (the grid core) with 8 blades (vanes), each of which is approximately 1 mm in thickness, around the center part (Fig. 2). The angle (α) of the blades from the horizontal plane is approximately 65° and the ratio of flow radii, inner

Table 1
Flow rates

		Equivalence ratio					
		0.5		0.6		0.75	
		Fuel	Air	Fuel	Air	Fuel	Air
Flow rates	Volumetric (L/s)	0.098	1.87	0.116	1.85	0.143	1.82
	Mass (g/s)	0.065	2.28	0.0776	2.22	0.096	2.18

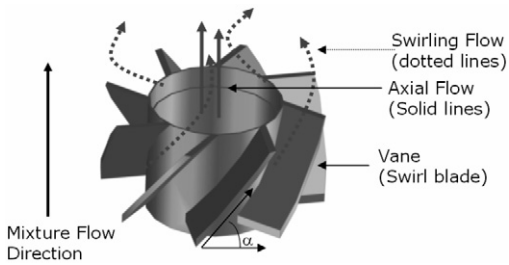

Fig. 2. Swirler.

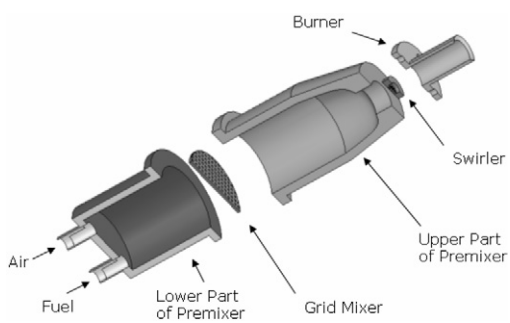

Fig. 3. Swirl burner assembly.

axial flow versus outer swirling flow, is 1/2. The premixer is 7.62 cm in diameter and 46 cm long and has a grid inserted between the upper and lower parts of the premixer to enhance fuel/air mixing.

Many other types of low-swirl burners use small air jets to swirl the premixed flow inside the burner tube which perturbs the core flow. This swirl-burner design leaves the center core flow undisturbed and produces a smooth divergent flow field above the burner exit. Flow divergence provides a highly effective means for stabilizing lean premixed combustion because it enables the flame to settle where the local velocity equals the flame speed [22]. This burner was also selected because other investigators have performed various studies on this burner and any results obtained herein can be compared to this existing body of work. Testing has included operation of this burner in an industrial gas turbine at realistic pressures and flow rates [23], lab scale PIV and PLIF measurements [22], and numerical modeling of similar burners [24].

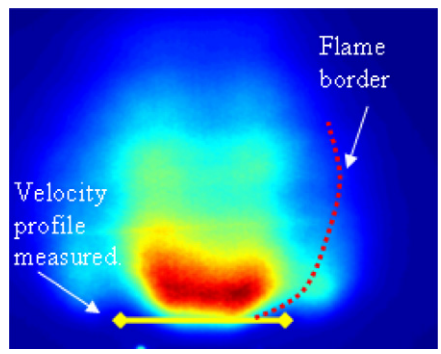

Fig. 4. Image of the time-averaged OH–PLIF signal.

3. Results

Culick [25] defines ΔE to be the energy transfer from the combustion process to the acoustic oscillations per cycle, which is given by

$$\Delta E = \frac{\gamma - 1}{\gamma} \int dV \int_{t}^{t+\tau_n} \frac{p'}{\bar{p}} Q' \, d\tilde{t}. \tag{1}$$

We use a nondimensional parameter that is defined as the Rayleigh index [12]:

$$R_{\mathrm{f}} = \int_{0}^{1} \frac{p' q'}{p_{\mathrm{rms}} \bar{q}} \, d\xi. \tag{2}$$

Fig. 5 shows an example of the heat release fluctuation, showing that the heat release rate fluctuates at the driving acoustic frequency. Here, the reference acoustic wave is shown, along with the heat release. Note that under the conditions for the data shown ($\phi = 0.6$, 85 Hz), the acoustic wave and the heat release fluctuation are approximately in phase, although there are significant phase shifts observed at other frequencies. The change in heat release rate is assumed to be proportional to the change in OH concentration.

3.1. Local Rayleigh index distribution

The Rayleigh index is calculated using Eq. (2) and a few selected local Rayleigh index distributions are shown in Figs. 6–8. The positive values indicate

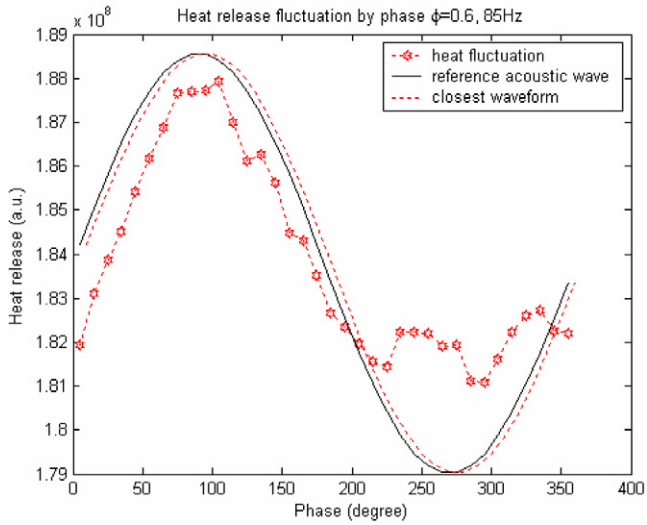

Fig. 5. Heat release fluctuation as a function of phase angle.

the region where the energy is positively fed back to the acoustic wave, thus magnifying the amplitude of the acoustic waves at that frequency, and a negative value indicates an acoustically dampening region.

Axial velocity distribution of a jet flow out of the swirl burner was measured by Cheng et al. [22] at locations indicated in Fig. 4 (that work used a swirl burner similar to the one employed here). Along the dotted line (the flame border on Fig. 4), the maximum velocity gradient is located so that strong shear mixing between the inner flow and the outer flow occur. In this shear-mixing region, it is expected that vortex structures form due to the differences in flow properties such as the axial velocity and combustion-induced density and baroclinic effects.

Figs. 6–8 show the circular patterns in the spatial distribution of the Rayleigh index in the region that are expected to be the shear mixing zone (from comparison with the axial velocity distribution [22]). These circular structures are believed to be slices of toroidal structures in 3-D. Each toroid is characterized by the same sign of Rayleigh index thus forming islands of similar tendency toward instability. It is in the leaner part of the flame that the toroidal structures are observed due to the shear entrainment of the outer flow (air in this case). Where the flow is lean, it is more susceptible to combustion instabilities because the values of the Rayleigh indices are larger.

For this system, it seems that the thermoacoustic coupling (Rayleigh index) is linked to shear-layer instability (Kelvin–Helmholtz). Vortex ring structures have been observed to grow in the jet boundary [26–29]. The driving force for formation of these vortical structures is the axial component of the fluctuating flow velocity due to acoustic excitations [26] and the shear layer formed with the outer flow. Ganapathisubramani et al. [30] performed PIV measurements of a cold-swirl-jet flow and showed that there exists an alternating circular pattern of vortical cores and straining region. Numerical simulations of Ghoniem et al. [29] showing that the unstable shear layer, rather than the acoustic field, acts as the thermoacoustic resonator. They report patterns similar to those observed in Figs. 6–8 in their shear regions.

The toroid structure is located in the shear layer, where the fuel/air mixture is mixed with the outer air flow. The mixture becomes even leaner and forms weaker heat release rate zones at the flame boundary. This would result in the Rayleigh distribution following the shape of the vortical distributions. Though the flame intensity is greater toward the center (Fig. 4), the central region is not much affected by the driving acoustic waves, in terms of the Rayleigh index (Figs. 6–8).

The global Rayleigh index (Fig. 9) has significant values only when the jet core region is thermoacoustically coupled such as at 210 Hz (bottom right, Figs. 6–8). The majority of the flow is the axial direction (hence “low” swirl), and thus most of the energy is released in the central region. The major contribution toward instability comes from the fuel jet core; the global Rayleigh index and flame response functions are significant only at frequencies at which the fuel jet core is thermoacoustically coupled to the imposed acoustic oscillations. At this conditions the contributions from the toroidal structures are likely to cancel each other.

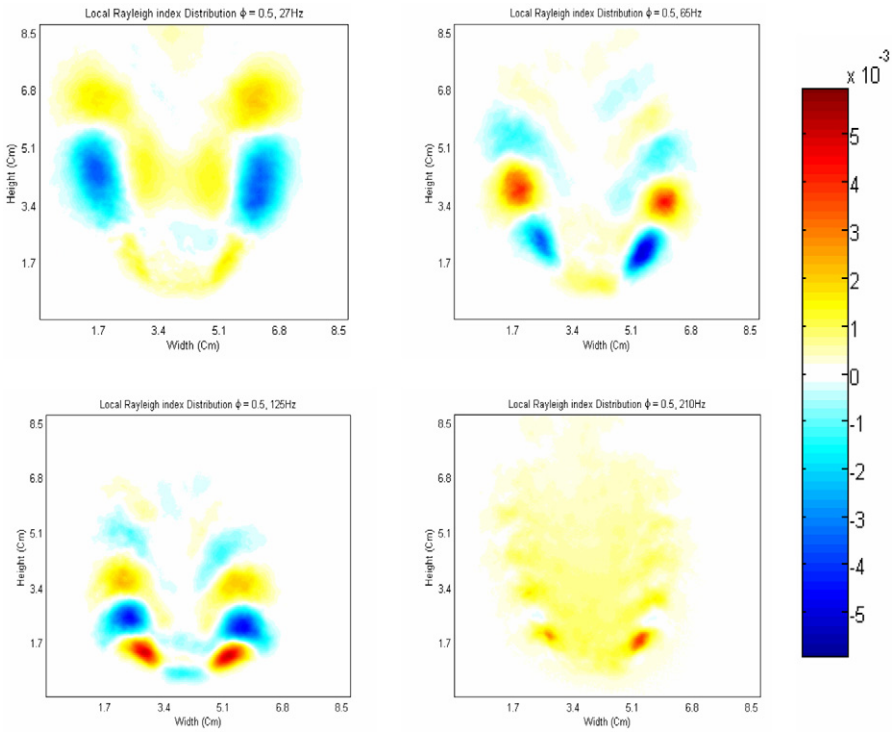

Fig. 6. Local Rayleigh index distribution for forcing frequencies of 27, 65, 125, and 210 Hz ($\phi = 0.5$).

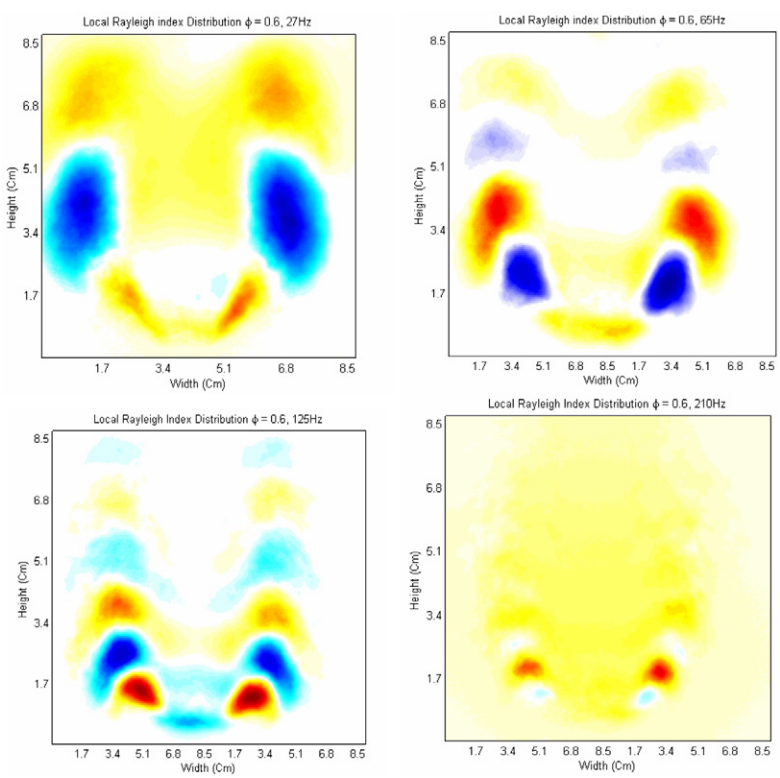

Fig. 7. Local Rayleigh index distribution for forcing frequencies of 27, 65, 125, and 210 Hz ($\phi = 0.6$).

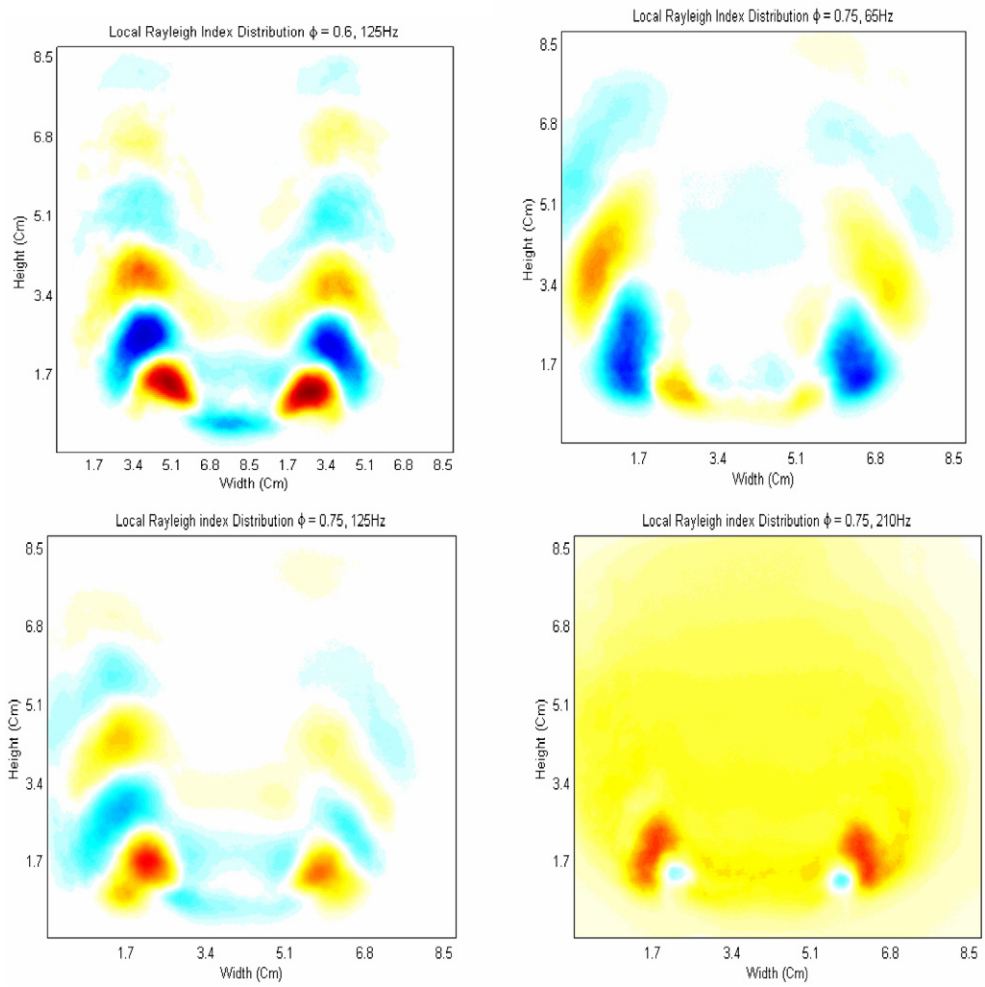

Fig. 8. Local Rayleigh index distribution for forcing frequencies of 27, 65, 125, and 210 Hz ($\phi = 0.75$).

The occurrence of the observed toroid structures is a strong function of excitation frequency, as is shown by Figs. 10–13. If these toroidal structures are actually shear-layer vortices (which must be verified by flow field measurements in the future), then the formation and growth of these vortices is highly sensitive to acoustic excitation. That is, the vortex structures created by the shear mixing are modulated by the imposed acoustic excitations and they appear to control the combustion response for this type of low-swirl burner.

The size of the toroidal structures becomes smaller and smaller until the acoustic excitation reaches a given frequency of about 100 Hz. The number of toroidal structures in the flame region is an indicator of the frequency of the toroid formation in the flow field. It does not appear that these structures are the result of vortex shedding, since they initially grow in strength before weakening downstream. As shown in Figs. 6–8, the spatial frequency of toroid occurrence increases with forcing frequency for all values of the equivalence ratio. Values of the local Rayleigh index were determined at locations along the peaks and troughs of the toroidal patterns (the dotted line in Fig. 4), and these values are presented in Figs. 10–12.

Figs. 10–12 show how the spatial distribution of the toroidal structure varies along the flame boundary in the shear layer. There are two main features: (1) the wave number of the fluctuation increases with frequency to some asymptotic value (approximately 35 m^{-1}, see Fig. 13); (2) the toroid-occurrence frequency is almost linearly proportional to the excitation frequency up to 100 Hz before it approaches an asymptotic value.

If the toroidal structure observed in our images of the local Rayleigh index is related to the vortex structures that are caused by the Kelvin–Helmholtz instability, then the increase in the wavenumber to

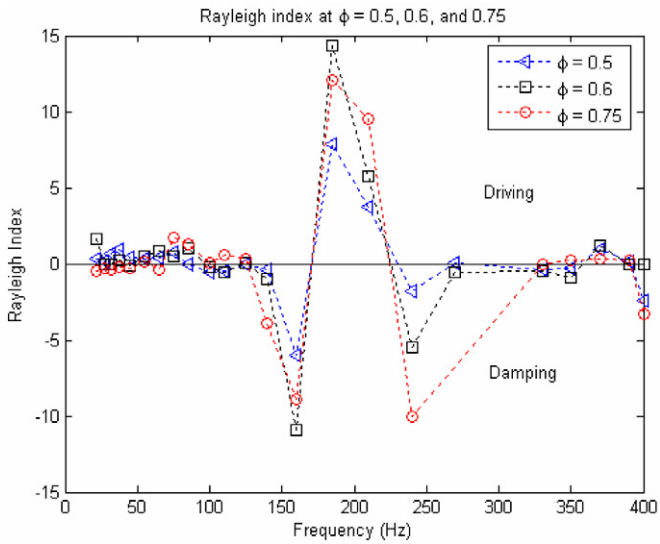

Fig. 9. Global Rayleigh index.

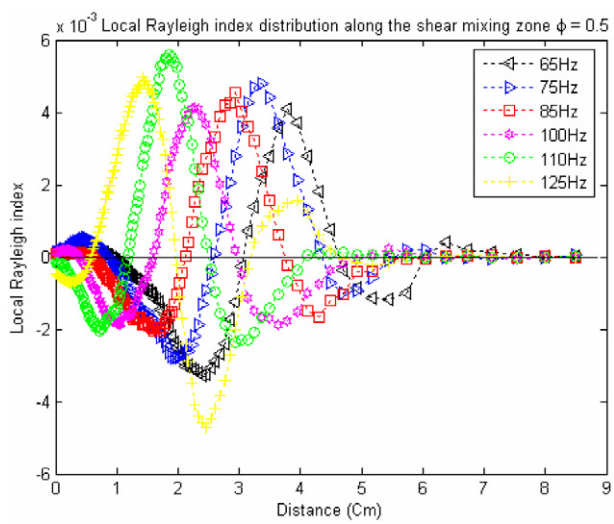

Fig. 10. Rayleigh index distribution along the flame boundary for $\phi = 0.5$.

a certain asymptotic value in Fig. 13 could be related to the phenomena of vortex shedding. The lower frequencies (100 Hz) seem to be associated with a vortex shedding frequency causing the trends seen in Fig. 13. The transition of the wave numbers from the low-frequency excitations (22–100 Hz) to the higher frequencies (>100 Hz) implies that the lock-on mode exists only in the low-frequency region. The trend is that the wavenumber does not seem to be linked to the particular forcing frequency for frequencies over 65 Hz, while the wavenumber and the forcing frequency are correlated at lower frequencies (<65 Hz).

3.2. Global Rayleigh index

The global Rayleigh index is a measure of the overall stability of the flame at a particular excitation frequency. This quantity is obtained by integrating the normalized index $\{p' \cdot q'\}/\{p_{\mathrm{rms}} \cdot q_{\mathrm{mean}}\}$ over the entire region that is defined by the present images. The variables q' and q_{mean} are calculated from the signal recorded at each pixel.

Fig. 9 shows the values of the global Rayleigh index, which indicates the frequency at which the flame is expected to be stable and the frequencies at which it is not. The measurements show that at frequencies of

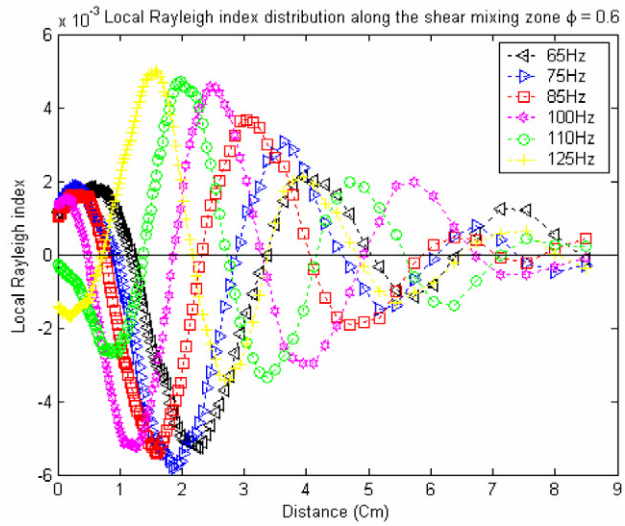

Fig. 11. Rayleigh index distribution along the flame boundary for $\phi = 0.6$.

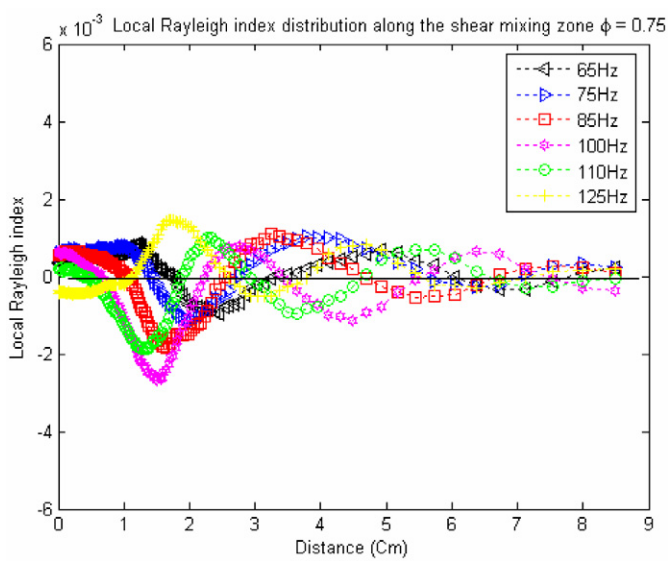

Fig. 12. Rayleigh index distribution along the flame boundary for $\phi = 0.75$.

150 to 250 Hz, there are local frequency ranges where significant damping (Ra < 0) and driving (Ra > 0) occur. It is also shown that the values recorded at the three different equivalence ratios (0.5, 0.6, and 0.75) agree with one another. This supports the assertion that the measurement method employed here is reliable and reproducible.

3.3. Global flame response function

The response function provides information about how the flame reacts to outside disturbances. Following Pun et al. [12], the response function is defined:

$$C_R = \frac{q'/\bar{q}}{p'/\bar{p}}. \tag{3}$$

The magnitude of this function is normally reported in decibels (dB), and the phase shift is measured from the phase of exciting acoustic waves. A positive value of phase shift mean that the peak of the heat release fluctuation is shifted to the right (to later times) than the peak of the acoustic waves.

Fig. 14 shows the magnitude and phase shift of the global response function integrated over the im-

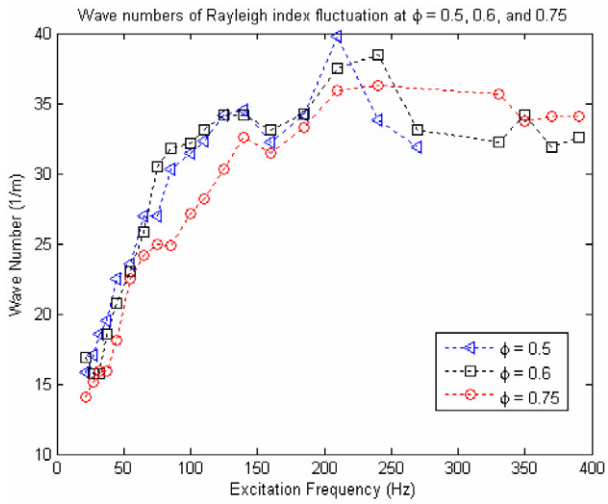

Fig. 13. Spatial frequency of toroidal structures.

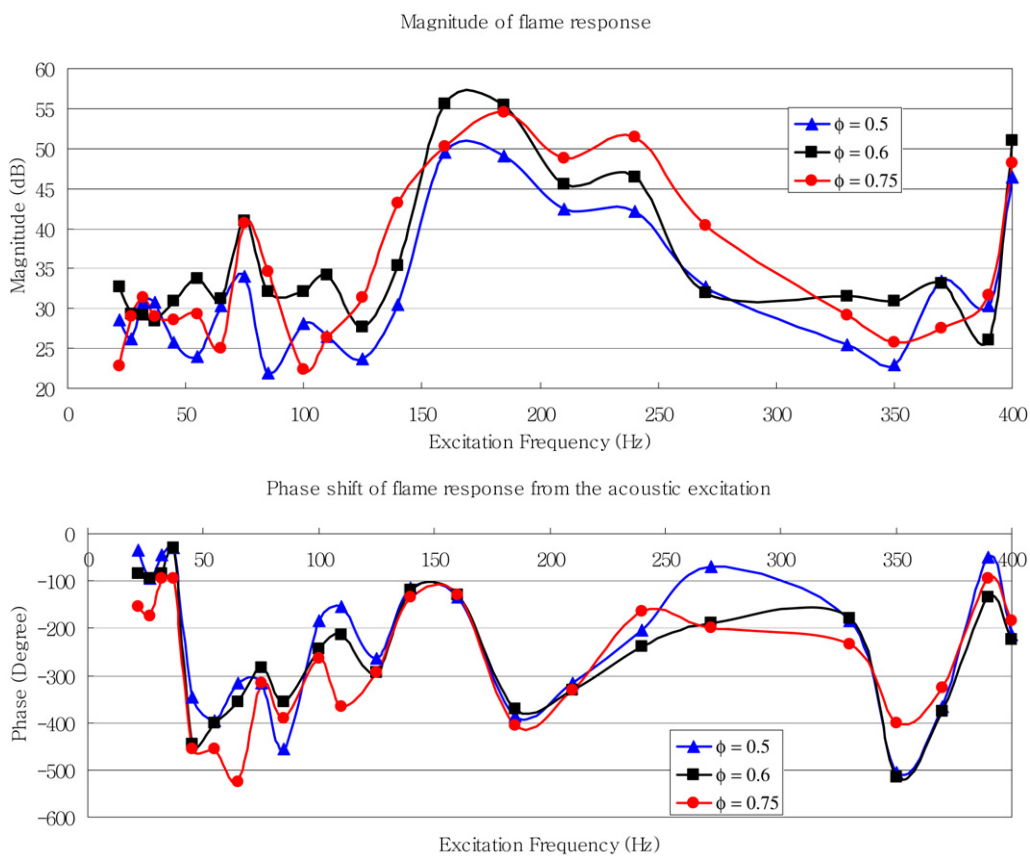

Fig. 14. Magnitude (top) and phase shift (bottom) of flame response function.

aged domain. For the three different equivalence ratios (0.5, 0.6, and 0.75), values of the response function are almost identical, and vary only slightly with excitation frequency. As the swirl is varied for three different equivalence ratios, the global quantities (the Rayleigh index and flame response function) are not affected significantly. This means that swirl causes a more stable and consistent performance of the sys-

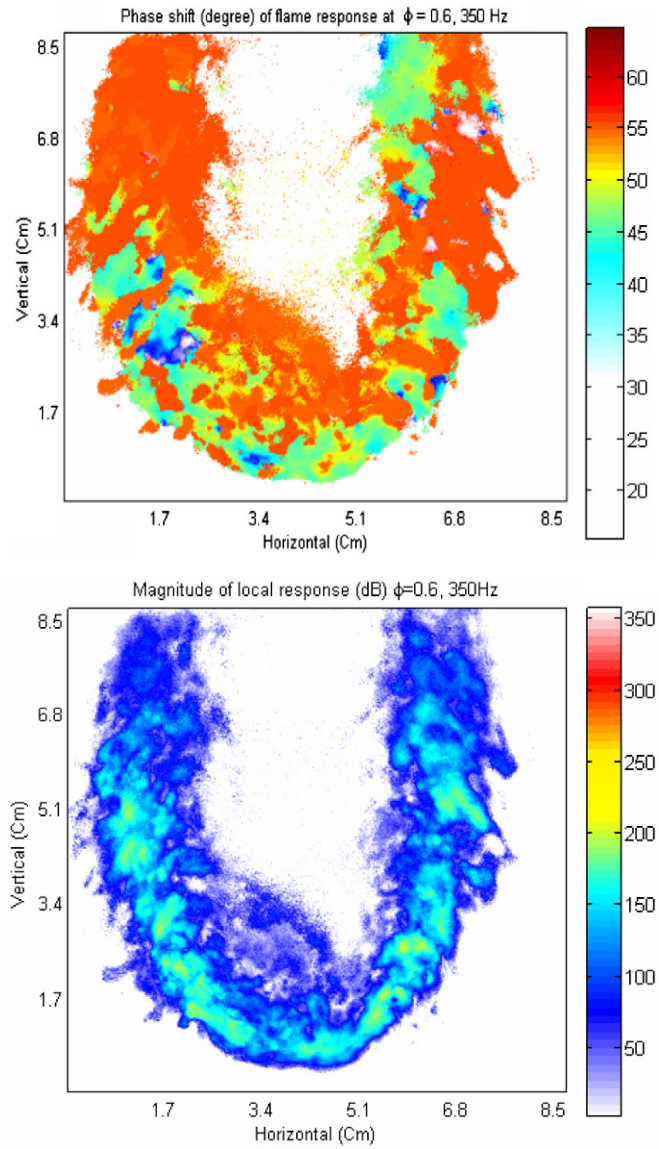

Fig. 15. Distribution of local flame response function ($\phi = 0.6$, 350 Hz); magnitude (top) and phase (bottom) with respect to the driving acoustic wave.

tem within the range of equivalence ratios studied. The phase shift shows significant frequency dependence, but negligible dependence on the equivalence ratio.

Using a basic acoustic analysis, if the present combustor geometry had a strong acoustic resonance, the frequency of the resonant standing wave would be approximately 120 Hz at a flame temperature of 1800 K. Such an acoustic resonance would be expected to create a sharp peak in the power spectrum, which was not measured. So the plateau in Fig. 9 that exists from approximately 150 to 250 Hz is not a combustor resonance mode. Instead it indicates the zone where the flame is thermoacoustically unstable, and hence much more sensitive to the outside disturbances than at other frequencies.

Comparing Figs. 9 and 14, the frequencies corresponding to large absolute global Rayleigh indices coincide with frequencies for which the flame response function is especially large. This is due to the fact that the Rayleigh index and the flame response function essentially quantify the same physical quantities (p' and q') though the way in which these quantities are coupled is not the same, as was shown by Eqs. (2) and (3). Thus it is of no surprise, based on the definition of the Rayleigh index, that the phase shift of the

flame response is close to zero (or 360°) at frequencies for which the Rayleigh index has a maximum value (e.g., at 185 Hz).

3.4. Distribution of local flame response function

Using Eq. (3), the local response function was calculated in the same manner as the global response. The primary difference is that instead of integrating over the entire imaged region, the values at each pixel are reported. The local maximum of the Rayleigh index coincides with the phase shift of 0° or 360°, while the local minimum coincides with a phase shift of approximately 180°, as was seen in Fig. 15.

Comparing the local Rayleigh index distribution and the spatial distribution of the magnitude of the flame response function, we conclude that (1) large values of the local response function are concentrated in the shear mixing zone, which is along the flame boundary, and (2) the local phase distribution coincides with local peaks (at 0° or 360°) and troughs (at 180°) of the Rayleigh index.

4. Conclusion

Two-dimensional images of the local Rayleigh index were measured using OH–PLIF in a low-swirl burner developed by Cheng. The Rayleigh index images were found to have a distinct feature: a series of toroidal structures was observed in the shear layer mixing zone. This indicates that thermoacoustic coupling is significant in the shear-layer region near the flame boundary. The coupling is believed to be due to the shear entrainment of ambient air, which dilutes the already lean initial fuel/air mixture and makes those regions more unstable. The toroidal structures become wider as the equivalence ratio is increased and appear to grow outward and extend to locations that are leaner and more susceptible to instabilities. The toroidal structures do not appear to be caused by the swirl created by the burner; if they were created by swirl they would be asymmetric and would be convected by the spiraling flow field. The structures are observed to be perfectly symmetric over all excitation frequencies and equivalence ratios. This leads to the conclusion that the driving force for the formation of this structure is not the swirling tangential and radial flows, but is the axial component of the flow field that exerts shear stress on the outer air flow. The toroidal structures appear to be closely coupled to the excitation frequencies. For this reason, the measured trends in the local and global flame response function, as acoustic forcing frequency is varied, are similar to the trends measured for the Rayleigh index.

Acknowledgments

This work was supported in part by the Air Force Office of Scientific Research (AFOSR) under Grant F49620-03-1-0384 (Dr. Mitat Birkan, Program Manager). Authors are grateful to Dr. Robert Cheng in Lawrence Berkeley Livermore Laboratory for letting us use the swirl burner and the premixer for our experiments. The authors also thank the University of Iowa and the California Institute of Technology for their support.

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